

Operational Amplifier

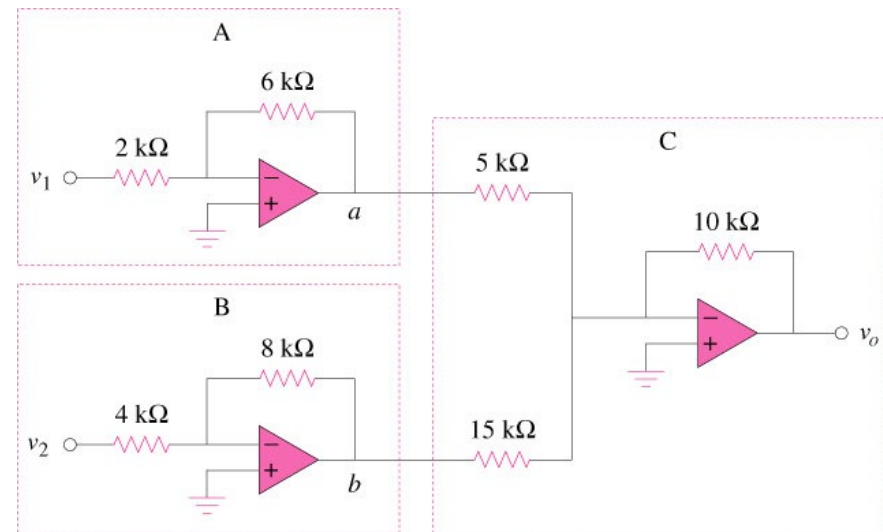
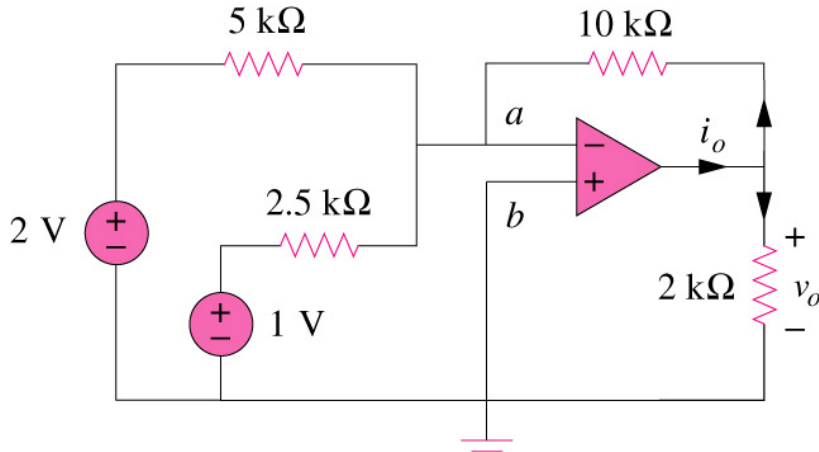
Video 10

Electrical Science

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Operational Amplifier

- Summing Op-Amp
- Difference Op-Amp
- Cascaded Op Amps
- Application: Digital-to Analog Converter



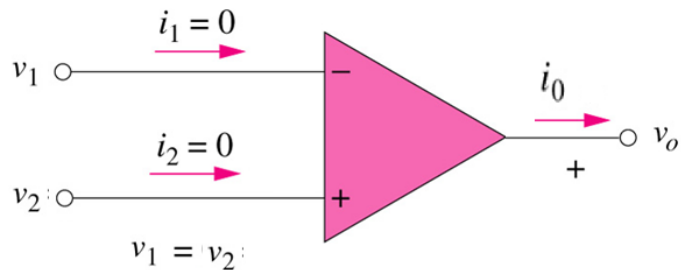
Op Amp Circuit Diagram



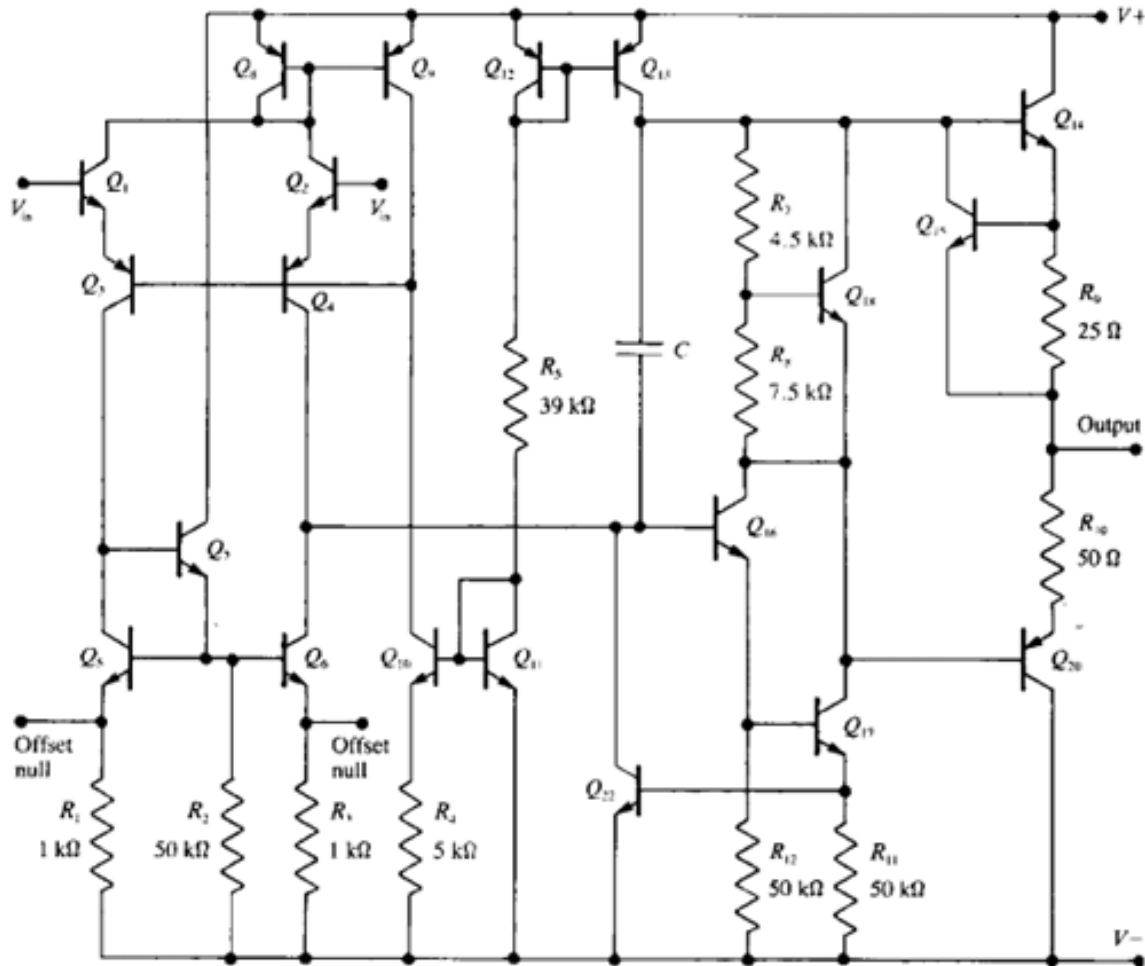
LM 741

Balance	1	8	No connection
Inverting input	2	7	V^+
Noninverting input	3	6	Output
V^-	4	5	Balance

Pin Configuration



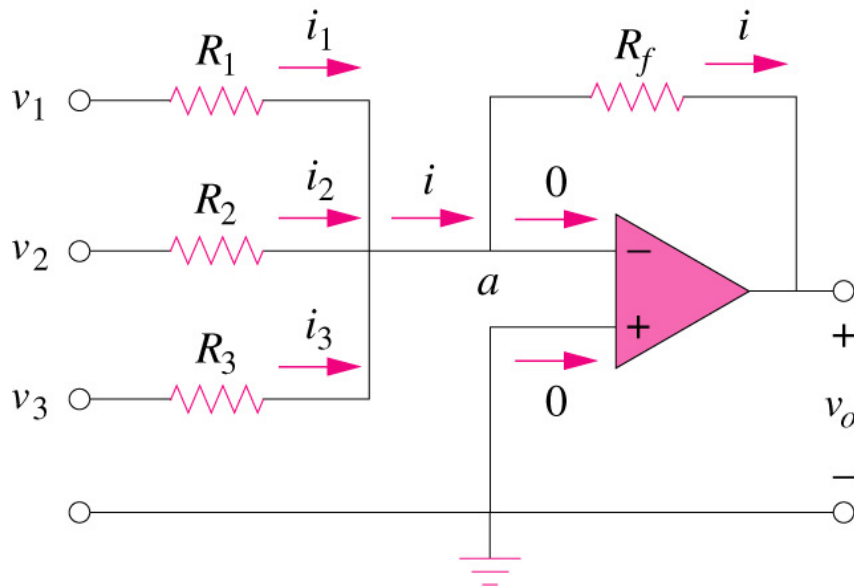
Ideal Op-Amp



20 Transistors, 11 Resistors, 1 Capacitor

Summing Amplifier

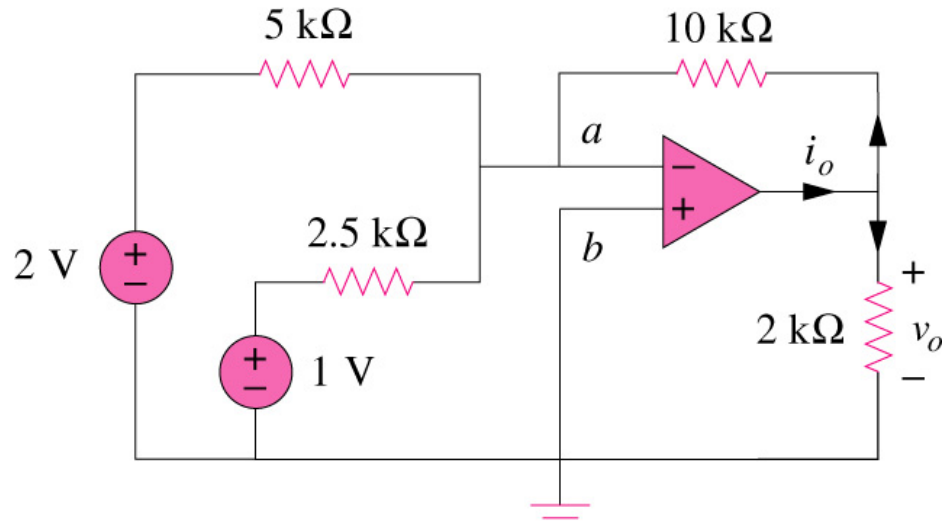
- Summing Amplifier is an op amp circuit that combines several inputs and produces an output that is the weighted sum of the inputs.



$$v_o = -\left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

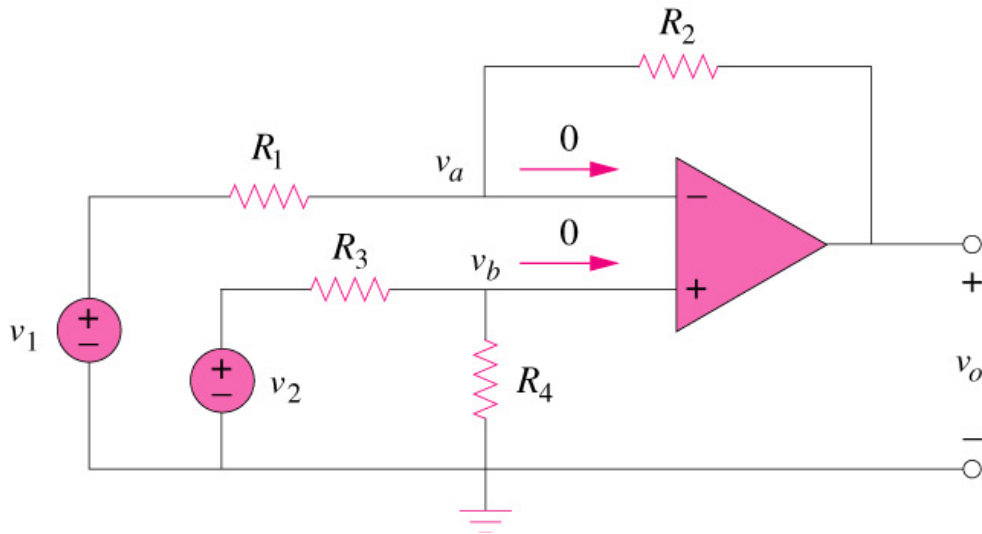
Summing Amplifier Example

Calculate v_o and i_o in the op amp circuit shown below.



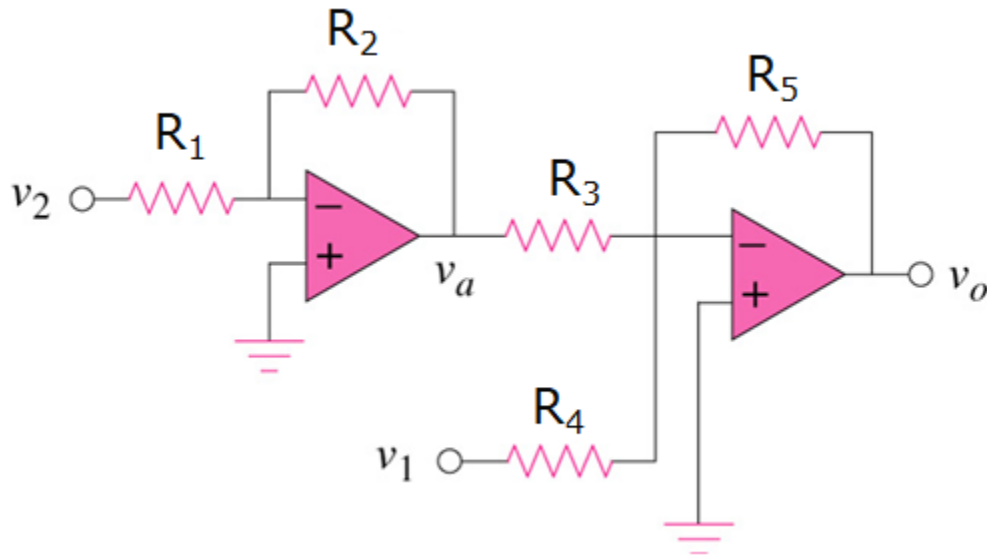
Difference Amplifier

- Difference amplifier is a device that amplifies the difference between two inputs but rejects any signals common to the two inputs.



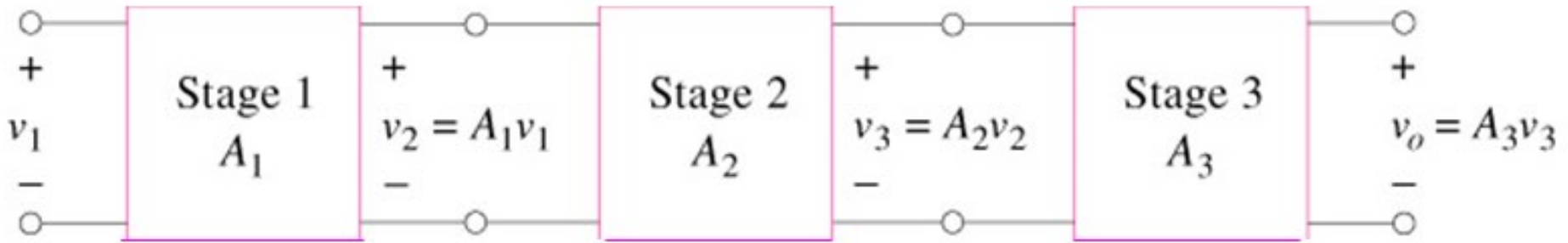
Difference Op-Amp Example

Determine R_1 , R_2 , R_3 , R_4 and R_5 so that $v_o = 3v_2 - 5v_1$ for the circuit shown below.



Cascaded Op Amp

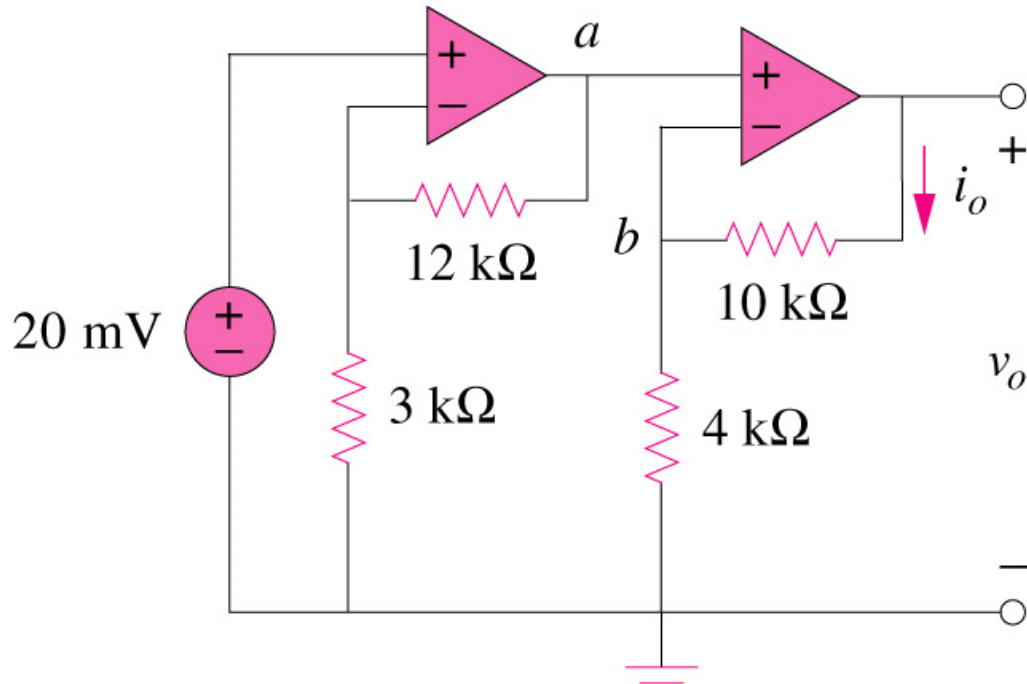
It is a head-to-tail arrangement of two or more op amp circuits such that the output to one is the input of the next.



$$v_o = A_1 A_2 A_3 v_1$$

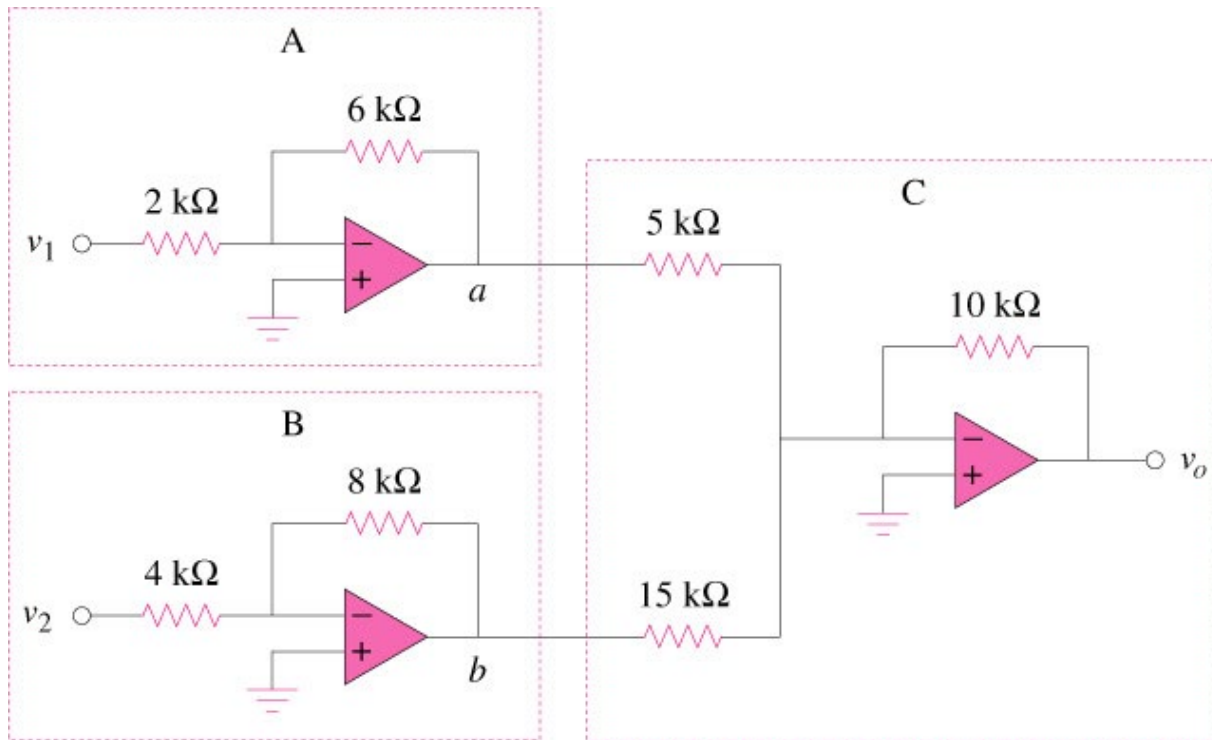
Cascaded Op Amp Example

Find v_o and i_o in the circuit shown below.



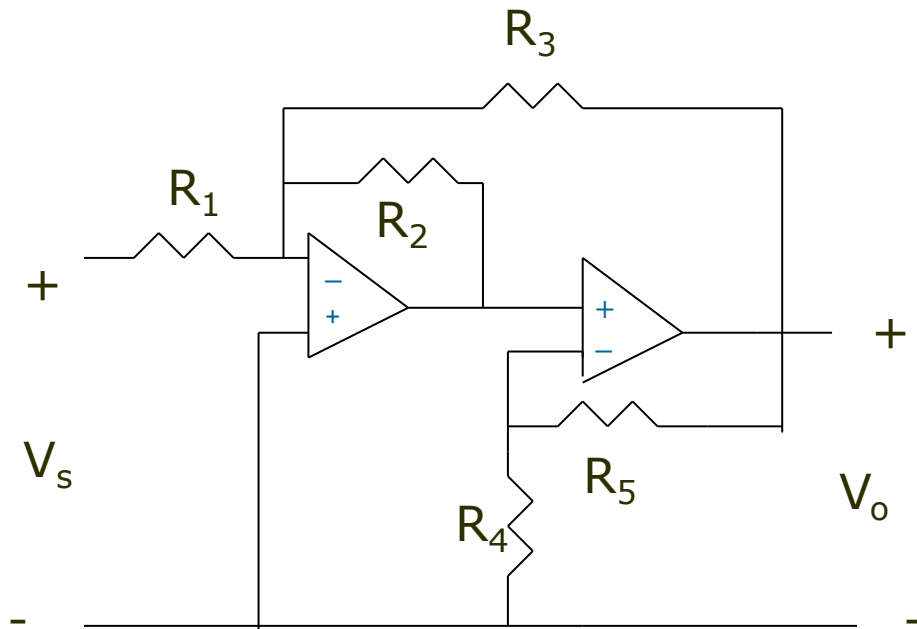
Cascaded and Parallel Op Amp

If $v_1 = 1V$ and $v_2 = 2V$, find v_o in the op amp circuit shown below.



Cascaded Op Amp Example

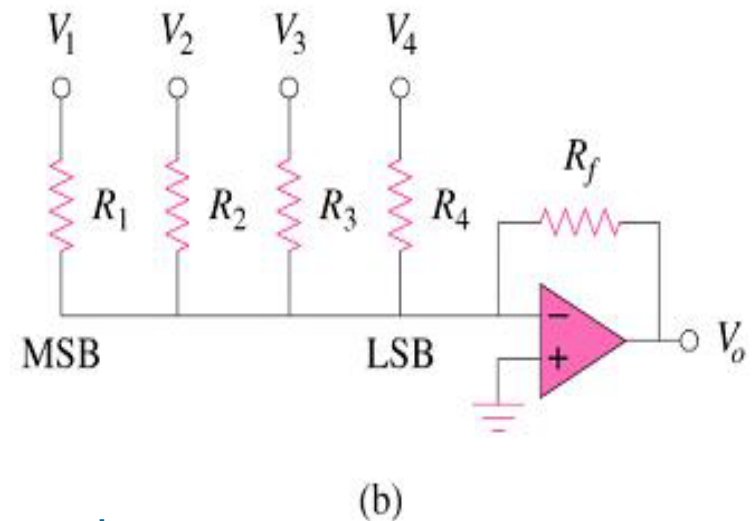
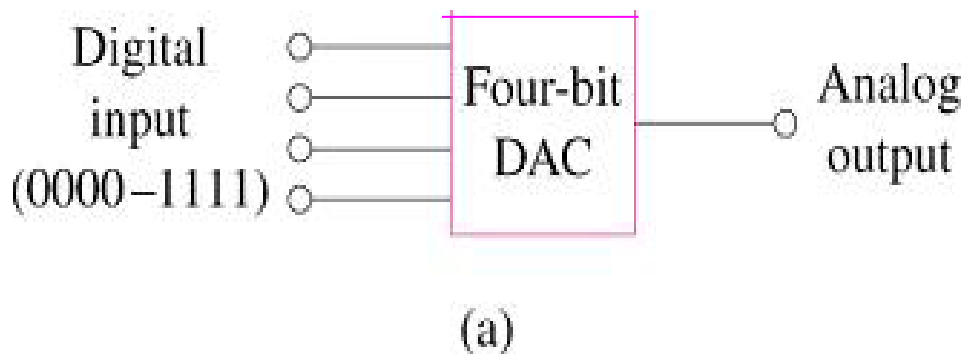
Determine the voltage transfer ratio v_o/v_s in the op amp circuit below.



Application (DAC)

- Digital-to Analog Converter (DAC) : it is a device which transforms digital signals into analog form.

Four-bit DCA: (a) block diagram (b) binary weighted ladder type



$$-V_o = \frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 + \frac{R_f}{R_4} V_4$$

where

V_1 – MSB, V_4 – LSB

V_1 to V_4 are either 0 or 1 V

Application (DAC) Example

For the circuit shown below, calculate v_o if $v_1=0V, v_2=1V$ and $v_3 = 1V$, equivalent to 3-bits binary (011).

